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Engineering,
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(PDEU)
Software Engineering(24CS209P)

Lab Experiment - 7

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Objective:

Start coding the project.

Basic Theory:

Clean code is easy to read and understand.

Git is used to manage code changes.

Lab Activities:

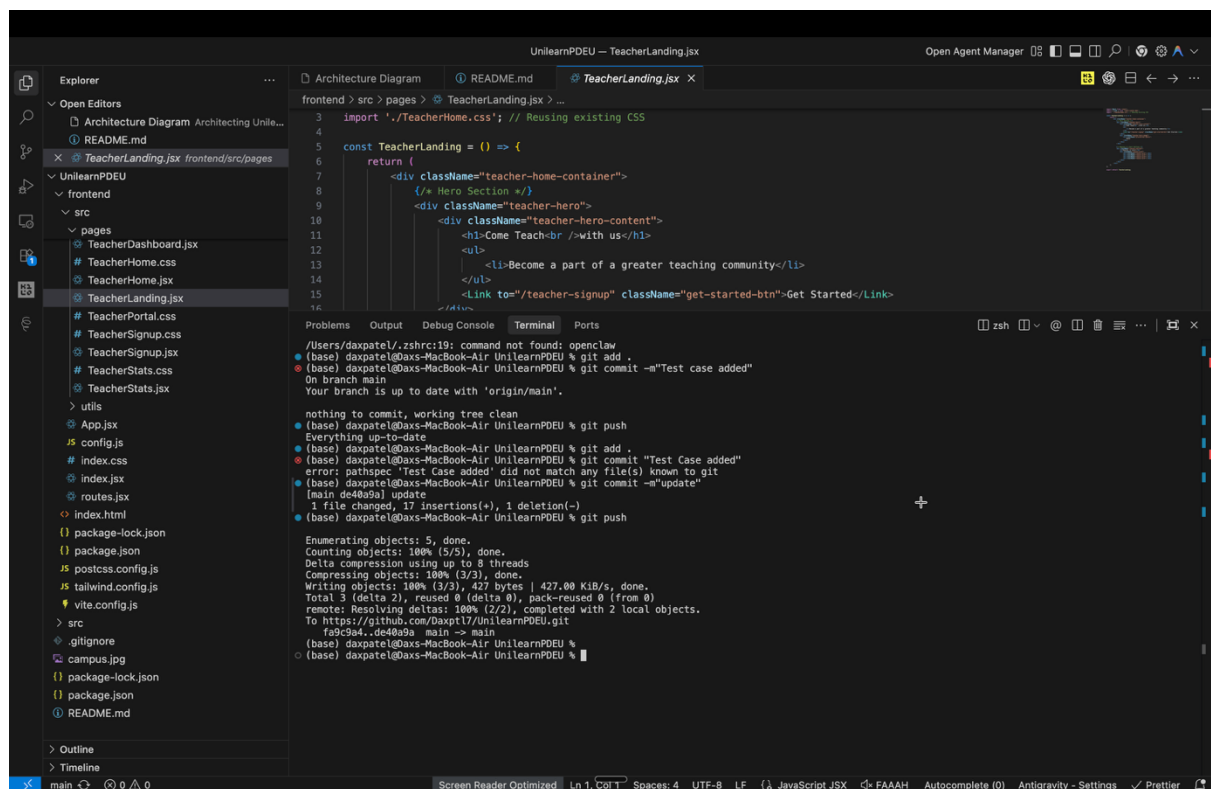
1. Create GitHub repository
2. Setup coding environment
3. Create class files
4. Implement basic features

Deliverables

1. Github repository link:

<https://github.com/Daxptl7/UnilearnPDEU.git>

2. IDE Screenshots:



3. Sample source code:

```
import React, { useState, useEffect } from 'react';
import { Link } from 'react-router-dom';
import TeacherDashboard from './TeacherDashboard';
import './Home.css';

const Home = ({ user, viewMode, toggleViewMode }) => {
  // State declarations
  const [recommended, setRecommended] = useState([]);
  const [loading, setLoading] = useState(true);
  const [opacity, setOpacity] = useState(0.3);

  // Teacher View Handler
  if (user?.role === 'teacher' && viewMode === 'teacher') {
    return <TeacherDashboard toggleViewMode={toggleViewMode} />;
  }

  useEffect(() => {
    const loadCourses = async () => {
      try {
        const { fetchPublicCourses } = await import('../api/course.api');
        const response = await fetchPublicCourses();
        if (response.success) {
          let allCourses = response.data;
          console.log("Fetched courses:", allCourses);

          // Filter by Student School if applicable
          if (user && user.role === 'student' && user.school) {
            console.log("Filtering for school:", user.school);
            const schoolCourses = allCourses.filter(c => c.category ===
user.school);

            setRecommended(schoolCourses);
          } else {
            setRecommended(allCourses);
          }
        }
      } catch (error) {
        console.error('Failed to load courses:', error);
      } finally {
        setLoading(false);
      }
    };
    loadCourses();
  }, []);

  // Scroll Opacity Effect
  useEffect(() => {
    const handleScroll = () => {
      const scrollFactor = Math.min(window.scrollY / 2000, 0.2);
```

```

setOpacity(0.3 + scrollFactor);
};
window.addEventListener('scroll', handleScroll);
return () => window.removeEventListener('scroll', handleScroll);
}, []);

return (
  <div className="home-wrapper">
    { /* HERO SECTION */ }
    <section
      className="heroContainer"
      style={{
        backgroundImage: `linear-gradient(rgba(0, 0, 0, ${opacity})),
        rgba(0, 0, 0, ${opacity}))`, url('/campus.jpg')`
      }}
    >
      <div className="heroContent">
        <h1 className="heroTitle">UniLearn | PDEU</h1>
        <p className="heroSubtitle">Empowering the future through
        Digital education and research.</p>
      </div>
    </section>

    { /* RECOMMENDED COURSES */ }
    <div className="section-container">
      <h3>
        {user && user.role === 'student' && user.school
          ? `Recommended for you (${user.school})`
          : "Recommended for you / Trending"}
      </h3>

      {loading ? (
        <p>Loading courses...</p>
      ) : (
        <div className="scroll-container">
          <div className="scroll-content">
            { /* Duplicate list for seamless loop (A + A) */ }
            {[...recommended, ...recommended].map((course, index)
=> (
              <Link key={`_${course._id}-${index}`}
to={`/_course/${course.slug}`} className="course-card" style={{
              minWidth: '280px', borderRadius: '12px',
overflow: 'hidden', boxShadow: '0 4px 10px rgba(0,0,0,0.1)', background: 'white',
display: 'block', color: 'inherit', flexShrink: 0
              }}>
                <div className="course-card-thumb" style={{
                  height: '160px',
                  backgroundImage:
`url(${course.thumbnail})`,
                  backgroundSize: 'cover',
                  backgroundPosition: 'center',

```

```

        backgroundColor: '#eee'
      }}</div>
      <div style={{ padding: '15px' }}>
        <span style={{ fontWeight: '600', fontSize:
'16px', display: 'block', marginBottom: '8px', overflow: 'hidden', textOverflow:
'ellipsis', whiteSpace: 'nowrap' }}>{course.name}</span>
        <span style={{ fontSize: '14px', color:
'#666' }}>By {course.instructor?.name || 'Instructor'}</span>
      </div>
    </Link>
  )})
</div>
</div>
  )}
</div>

{/* STATS SECTION (Placeholder for IdeaLab2 Stats) */}
<div className="section-container" style={{ background: '#f9f9f9',
textAlign: 'center' }}>
  <h3 style={{ borderLeft: 'none' }}>Our Impact</h3>
  <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-
fit, minmax(200px, 1fr))', gap: '40px', marginTop: '40px' }}>
    <div>
      <h2 style={{ fontSize: '48px', color: '#A6192E',
fontWeight: '800' }}>50+</h2>
      <p>Lab Equipment</p>
    </div>
    <div>
      <h2 style={{ fontSize: '48px', color: '#A6192E',
fontWeight: '800' }}>200+</h2>
      <p>Projects Completed</p>
    </div>
    <div>
      <h2 style={{ fontSize: '48px', color: '#A6192E',
fontWeight: '800' }}>1000+</h2>
      <p>Students Trained</p>
    </div>
  </div>
</div>
  </div>
);
};

export default Home;

```